

## OSTROV UEDINENIJA, Russian Federation

WMO: 202740

Lat: 77.511N

Lon: 82.233E

Elev: 75

StdP: 14.66

Time Zone: 7.00 (E07)

Period: 86-96

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -36.6      | -34.2 | -41.1                         | 0.5 | -36.3 | -38.4 | 0.6 | -33.9 | 36.8                  | 1.3  | 34.2 | -2.3 | 10.5                  | 140  | 1.221 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 3.7                    | 39.5            | 38.6 | 37.4 | 36.6 | 35.9 | 35.1 | 38.7                | 39.3 | 36.8 | 37.3 | 35.3 | 35.7 | 15.3                 | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 38.0                            | 33.7 | 38.9 | 36.1 | 31.2 | 37.1 | 34.6 | 29.4 | 35.5 | 14.7          | 39.6 | 13.8 | 37.5 | 13.2 | 35.8 | 45.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 34.7          | 30.2 | 27.0 | DB                         | -42.6 | 45.3               | 5.0 | 3.2                                                | -46.3 | 47.6       | -49.2 | 49.4       | -52.0 | 51.2       | -55.7 | 53.5 | (4) |
| (5)               |      |      | WB                         | -42.7 | 41.9               | 5.0 | 2.0                                                | -46.3 | 43.4       | -49.2 | 44.6       | -52.0 | 45.7       | -55.6 | 47.1 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.4    | -15.6 | -14.9 | -11.2 | -2.3  | 16.4  | 28.8  | 33.0  | 31.9  | 26.9  | 11.8  | -5.1  | -12.0 |
|                                                                                  | DBStd       | 21.25  | 11.82 | 12.74 | 14.70 | 11.93 | 7.40  | 3.81  | 2.43  | 2.96  | 5.49  | 12.05 | 13.35 | 11.63 |
|                                                                                  | HDD50       | 15534  | 2034  | 1817  | 1897  | 1570  | 1041  | 635   | 527   | 562   | 693   | 1184  | 1652  | 1922  |
|                                                                                  | HDD65       | 21009  | 2499  | 2237  | 2362  | 2020  | 1506  | 1085  | 992   | 1027  | 1143  | 1649  | 2102  | 2387  |
|                                                                                  | CDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 13.5   | 13.3  | 13.1  | 13.6  | 13.7  | 12.6  | 12.3  | 12.7  | 13.0  | 13.9  | 15.8  | 12.6  | 14.7  |
| Precipitation                                                                    | PrecAvg     | 6.2    | 0.3   | 0.3   | 0.3   | 0.2   | 0.3   | 0.5   | 1.0   | 1.2   | 0.9   | 0.6   | 0.3   | 0.4   |
|                                                                                  | PrecMax     | 9.0    | 0.8   | 1.1   | 1.3   | 0.8   | 0.9   | 1.3   | 2.2   | 2.7   | 2.3   | 1.4   | 0.7   | 1.2   |
|                                                                                  | PrecMin     | 1.9    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.3   | 0.1   | 0.0   | 0.1   |
|                                                                                  | PrecStd     | 2.0    | 0.2   | 0.3   | 0.3   | 0.2   | 0.2   | 0.3   | 0.6   | 0.7   | 0.4   | 0.4   | 0.2   | 0.3   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 25.6  | 26.7  | 29.5  | 31.1  | 32.5  | 38.5  | 43.1  | 41.4  | 36.9  | 32.2  | 29.3  | 28.8  |
|                                                                                  |             | MCWB   | 24.9  | 26.1  | 29.2  | 30.4  | 32.2  | 37.0  | 40.7  | 40.9  | 36.2  | 31.3  | 29.0  | 27.9  |
|                                                                                  | 2%          | DB     | 19.4  | 19.2  | 25.4  | 26.9  | 31.8  | 35.1  | 40.2  | 39.3  | 34.8  | 31.3  | 25.6  | 22.0  |
|                                                                                  |             | MCWB   | 19.1  | 18.7  | 24.8  | 26.1  | 31.2  | 33.7  | 38.8  | 38.9  | 34.2  | 30.7  | 25.2  | 21.3  |
|                                                                                  | 5%          | DB     | 10.1  | 11.5  | 21.1  | 21.0  | 29.8  | 33.7  | 37.9  | 37.2  | 33.9  | 30.2  | 21.8  | 13.7  |
|                                                                                  |             | MCWB   | 9.8   | 10.9  | 20.8  | 20.2  | 28.9  | 32.7  | 36.9  | 36.5  | 33.4  | 29.4  | 21.3  | 13.2  |
|                                                                                  | 10%         | DB     | 0.9   | 5.4   | 11.9  | 16.2  | 27.0  | 33.0  | 36.4  | 35.7  | 32.9  | 28.3  | 15.8  | 5.0   |
|                                                                                  |             | MCWB   | 0.4   | 5.1   | 11.5  | 15.4  | 26.1  | 32.2  | 35.6  | 35.1  | 32.3  | 27.1  | 15.3  | 4.5   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 24.9  | 26.3  | 29.2  | 30.7  | 32.3  | 37.3  | 41.1  | 41.2  | 36.4  | 32.1  | 28.8  | 28.2  |
|                                                                                  |             | MCDB   | 25.5  | 26.5  | 29.4  | 31.2  | 32.3  | 38.5  | 42.2  | 41.2  | 36.8  | 32.1  | 29.0  | 28.6  |
|                                                                                  | 2%          | WB     | 18.9  | 18.8  | 25.1  | 26.3  | 31.2  | 34.1  | 38.9  | 38.9  | 34.4  | 30.9  | 25.2  | 21.6  |
|                                                                                  |             | MCDB   | 19.4  | 19.2  | 25.4  | 27.2  | 31.5  | 34.6  | 40.0  | 39.1  | 34.7  | 31.1  | 25.5  | 22.1  |
|                                                                                  | 5%          | WB     | 9.6   | 11.1  | 20.6  | 20.4  | 29.0  | 33.1  | 37.2  | 36.8  | 33.4  | 29.6  | 21.4  | 13.2  |
|                                                                                  |             | MCDB   | 10.0  | 11.6  | 21.1  | 21.1  | 29.7  | 33.5  | 37.9  | 37.1  | 33.6  | 30.0  | 21.8  | 13.7  |
|                                                                                  | 10%         | WB     | 0.5   | 5.0   | 11.5  | 15.5  | 26.2  | 32.4  | 35.7  | 35.2  | 32.5  | 27.0  | 15.2  | 4.3   |
|                                                                                  |             | MCDB   | 1.0   | 5.4   | 11.8  | 16.4  | 27.0  | 32.7  | 36.3  | 35.6  | 32.8  | 28.2  | 15.8  | 4.8   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.0  | 10.6  | 12.0  | 10.6  | 6.4   | 3.7   | 3.7   | 3.5   | 4.5   | 7.8   | 10.9  | 9.0   |
|                                                                                  |             | MCDBR  | 18.9  | 22.5  | 19.1  | 11.0  | 5.9   | 3.8   | 6.5   | 4.6   | 4.0   | 8.1   | 19.1  | 17.1  |
|                                                                                  |             | MCWBR  | 18.8  | 22.9  | 19.0  | 10.2  | 5.8   | 3.3   | 5.2   | 4.4   | 3.9   | 8.3   | 19.2  | 17.0  |
|                                                                                  | 5% WB       | MCDBR  | 18.5  | 23.0  | 19.0  | 9.6   | 6.1   | 3.9   | 6.3   | 4.5   | 3.9   | 8.9   | 19.4  | 16.9  |
|                                                                                  |             | MCWBR  | 18.5  | 22.8  | 19.1  | 9.7   | 6.2   | 3.4   | 5.1   | 4.4   | 4.1   | 9.2   | 19.5  | 16.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | N/A    | 0.182 | 0.258 | 0.292 | 0.315 | 0.311 | 0.334 | 0.328 | 0.284 | 0.353 | N/A   | N/A   |       |
|                                                                                  | taud        | N/A    | 1.845 | 1.934 | 1.992 | 2.030 | 2.212 | 2.376 | 2.471 | 2.345 | 2.704 | N/A   | N/A   |       |
|                                                                                  | Ebn at Noon | 0      | 20    | 170   | 235   | 253   | 264   | 252   | 233   | 183   | 29    | 0     | 0     |       |
|                                                                                  | Edh at Noon | 0      | 2     | 19    | 32    | 37    | 33    | 27    | 21    | 14    | 3     | 0     | 0     |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0      | 12    | 326   | 1126  | 1733  | 1883  | 1365  | 720   | 255   | 34    | 0     | 0     |       |
|                                                                                  | RadStd      | 0      | 2     | 33    | 82    | 97    | 155   | 173   | 105   | 43    | 5     | 0     | 0     |       |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page